KAI WANG | 王凯

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RESEARCH

- **Galaxy Evolution**: *What drives galaxy quenching, metal enrichment, and morphology transformation?*
- **Galaxy-Halo Connection**: *To what extent are the properties of galaxies shaped by their dark matter halos?*
- **Dark Matter Halo**: *How do halo structure, assembly history, and spatial distribution interplay with each other?*

EXPERIENCE

- Postdoctoral Research Associate, ICC and CEA, **Durham University** since 07/2024
- KIAA Fellow, Kavli Institute for Astronomy and Astrophysics, **Peking University** 07/2022 - 06/2024

EDUCATION

- Ph.D. in Astronomy, **Tsinghua University** 09/2017 - 07/2022
- Visiting Scholar, **University of Massachusetts, Amherst** 11/2019 - 10/2021
- B.S. in Astronomy, **University of Science and Technology of China (USTC)** 09/2013 - 07/2017

PUBLICATION SUMMARY

✦**48** publications (**36** since 2023); **20** as the first/corresponding author (**16** since 2023)
✦**>530** citations (**>425** since 2023); *H*-index: **>15**; open in NASA/ADS

SELECTED PUBLICATIONS

1. **Inflow-driven galaxy evolution - I. Revealing the physics of the fundamental metallicity relation**

   Kai Wang†, Carlton Baugh, N. F. Boardman, Sownak Bose, Zheng Cai, Shaun Cole, Carlos S. Frenk, Cedric Lacey, Peder Norberg, Yingjie Peng, Isabel Santos-Santos, Amélie Saintonge, Tom Theuns, Enci Wang
   submitted to MNRAS (2026) [arXiv: 2608.04784]

   I derived a unified framework for the chemical evolution of galaxies from the two mass-continuity equations, providing the first self-consistent explanation of the fundamental metallicity relation (FMR) and its redshift invariance to z~3. The FMR is a projection of the gaseous FMR, and its parameterisation encodes how star formation efficiency varies with stellar mass and redshift, and its shape shows galaxies evolve from inflow-driven to equilibrium.

2. **Inflow-driven galaxy evolution - II. A hierarchy in the scaling relations of star-forming galaxies**

   Kai Wang†, Carlton Baugh, Sownak Bose, Shaun Cole, Carlos S. Frenk, Cedric Lacey, Julio F. Navarro, Kyle A. Oman, Amélie Saintonge, Tom Theuns
   submitted to MNRAS (2026) [arXiv: 2608.04784]

   I identified a hierarchy among the three primary scaling relations of star-forming galaxies: the star-forming main sequence (SFMS), the gas-to-stellar mass relation, and the mass–metallicity relation (MZR). Halo formation time sets the SFMS; combined with the star formation efficiency, the SFMS gives the gas-to-stellar mass relation; and combined with the mass-loading factor, the gas-to-stellar mass relation gives the MZR. This hierarchy also naturally explains three higher-dimensional relations: the fundamental metallicity relation (FMR), its gaseous counterpart (gFMR), and the fundamental plane of star formation.

3. **Forged in Quenching: Morphological Transformation across Star-forming and Quiescent Galaxies in EAGLE**

Kai Wang†, Carlton Baugh, Sownak Bose, Shaun Cole, Carlos S. Frenk, Hao Fu, Cedric Lacey, Shengdong Lu, Aaron Ludlow, Peder Norberg, Yingjie Peng, Katy L. Proctor, Isabel Santos-Santos, Francesco Shankar, Tom Theuns, Enci Wang, Tao Wang, Vivienne Wild

submitted to MNRAS (2026) [arXiv: 2609.16187]

Using EAGLE, I showed that morphology actively drives SMBH growth and quenching central galaxies, rather than being passively inherited through progenitor bias. I also showed that merger-driven SMBH growth in semi-analytic models cannot reproduce the size difference between star-forming and quiescent galaxies, pointing to the need for a secular, morphology-dependent SMBH growth channel.

4. **An efficient and robust method to estimate halo concentration based on the method of moments**

Kai Wang†, Houjun Mo, Yangyao Chen, Joop Schaye

MNRAS 527, 10760 (2023) [arXiv: 2310.00200]

I proposed a robust moment-based estimator of halo concentration, now widely adopted in the simulation processing tools, like SOAP and AHF, and in the cosmological simulations, like FLAMINGO and COLIBRE.

5. **Dissect two-halo galactic conformity effect: The dependence of star formation activities on the large-scale environment for central galaxies**

Kai Wang†, Yingjie Peng†, Yangyao Chen

MNRAS 523, 1268 (2023) [arXiv: 2304.06886]

I found that backsplash galaxies explain the dependence of central galaxy quenching on large-scale environment and give rise to two-halo galactic conformity in both IllustrisTNG and SDSS.

GRANTS

- KIAA fellow start-up research funding (50,000CNY) 07/2022 - 07/2024
- China Scholarship for the Visiting Scholar ($45,600) 11/2019 - 10/2021
- National Astronomy Training Base (20,000CNY) 06/2016 - 06/2017
- National Astronomy Training Base (20,000CNY) 05/2015 - 05/2016

MENTORSHIP/SUPERVISION

- **Oscar Hickman**, Postgraduate student at Durham University since 02/2026
  Less is More: Unbiased Galaxy Clustering from N-Body Merger Trees
- **Sienna Folkes-Miller**, L4 student at Durham University since 09/2026
  Tracing the Invisible: How Galaxies Reveal the Dark Universe
- **Thomas Power**, L4 student at Durham University 10/2025 - 04/2026
  The Dance of Milky Way and Andromeda: A Celestial Ballet Across Cosmologies
- **Thomas Richardson**, L4 student at Durham University 10/2025 - 04/2026
  Where Shadows Follow Light: Baryon-Induced Dark Matter Halo Contraction in Hydrodynamical Simulations
- **Zeyu Gao**, graduate at Peking University 11/2022 - 07/2024
  Decoding the SEDs of galaxies with a prior from hydrodynamical simulations [arXiv: 2408.07749]
- **Chengyu Ma**, graduate at USTC 12/2023 - 07/2024
  Revisiting the fundamental metallicity relation with observation and simulation [arXiv: 2407.21716]
- **Haochen Jiang**, undergraduate at USTC 12/2023 - 10/2025
  Dissecting the quenching of massive central galaxies in TNG [arXiv: 2510.24886]
- **Xunda Sun**, graduate at the University of Chinese Academy of Sciences 06/2023 - 10/2024
  Characterizing the spatial distribution of the metal content for galaxies in FIRE2 [arXiv: 2409.09290]
- **Zhijun Zhang**, undergraduate at Peking University 09/2022 - 06/2023
  Identify protoclusters from high-redshift photometric surveys [Bachelor Thesis]

TEACHING

- Level-1 Physics Tutorial — Tutor, Durham University, 2024-2025
  Year-long weekly tutorials for five undergraduates (mechanics, E&M, quantum mechanics etc.), taught interactively with board work by students. Peer-observed by the Director of Education, Prof. Michael Hunt: "a well planned and conducted tutorial", noting the practice of "going beyond the model answer to explain underlying concepts".
- ExGal Summer School on Astrophysical Simulations — Tutorial assistant, Durham University, 2025
- Cosmology and Galaxy Evolution — Guest Lecturer, Peking University, Autumn 2023
- Observational Cosmology — Teaching Assistant, Tsinghua University, Autumn 2017
- Particle Cosmology — Teaching Assistant, USTC, Spring 2017
- General Relativity — Teaching Assistant, USTC, Autumn 2016

OUTREACH

- **Summer Science Exhibition 2026**, The Universe Makers Team — Royal Society, London, 2026
  The Summer Science Exhibition is the Royal Society's flagship annual public engagement festival celebrating the cutting edge of UK research and innovation. Welcoming over 10,000 total visitors—including 2,000 secondary school students and teachers—the high-capacity, six-day event features 13 interactive flagship exhibits and over 60 specialised talks and panels delivered by 100+ world-leading speakers.
- **Undergraduate Open Day**, Demonstrator — Durham University, 2026
  The Undergraduate Open Day is Durham University's flagship two-day recruitment event for prospective applicants and parents, showcasing its top-5 UK standing. Spanning 16 undergraduate colleges and dozens of departments, the event utilises a massive staff and student ambassador network to deliver presentations, exhibitions, and campus tours.
- **Schools' Science Festival**, Demonstrator — Durham University, 2025
  The Durham University Schools' Science Festival is an annual outreach program designed to inspire Year 9 and Year 10 students Durham University Schools' Science Festival 2025. The 2025 festival welcomed over 700 pupils from 22 schools across the region to participate in interactive, hands-on scientific projects.
- **Celebrate Science**, Demonstrator — Palace Green, Durham, 2024
  Celebrate Science is Durham University's annual public festival aimed at children aged 7–11, families, and school groups, designed to make complex research accessible. The event attracts over 5,000 to 7,500+ attendees annually, featuring over 200 volunteers to deliver interactive STEM workshops.

SERVICE

- **Referee** for MNRAS (since 2024), ApJ (since 2023), and A&A (since 2022)
- **LOC member** of "Looking into the dark: Frenk@75" — 2026
  The Looking into the Dark: Frenk@75 conference, held at the Royal Society in London, celebrated the 75th birthday of Professor Sir Carlos S. Frenk CBE FRS, recognising his contributions to cosmology and galaxy formation and his lasting influence on the students, postdoctoral researchers, and collaborators he has worked with throughout his career.
- **SOC member** and **discussion panelist** of session in NAM 2026 — 2026
  The NAM 2026 session Galaxy Chemistries Across Cosmic Time featured 2 invited talks, 10 contributed talks, 7 poster presentations, and a panel discussion, bringing together observers and theorists to discuss galaxy chemical evolution from the local Universe to the highest redshifts.
- **SOC member** of session "The Golden Era of Gravitational Lensing" of NAM 2025 — 2025
  The NAM 2025 session The Golden Era of Gravitational Lensing featured 18 contributed talks and 4 poster presentations, covering strong and weak lensing, microlensing, gravitational-wave lensing, dark matter, cosmology, and next-generation surveys including Euclid, JWST, Rubin LSST, Roman, and SKA.
- **Organiser** of CosmICConnections at Durham University — 2026
  CosmICConnections is a hybrid seminar and discussion series held every four weeks, providing an informal forum for local and visiting researchers to present their work, exchange ideas, and foster collaborations across astronomy and cosmology.

- **Organiser** of Friday Lunchtime Astronomy Talks (FLAT) at Durham University 2025-2026

  The Friday Lunchtime Astronomy Talk (FLAT) is a weekly seminar series featuring around 60 talks annually by local and visiting early-career researchers, fostering scientific discussion and collaboration among students and postdoctoral researchers.
- **LOC member** of National Astronomy Meeting (NAM) 2025 at Durham University 2025

  NAM 2025, hosted by Durham University, brought together over 1,400 attendees across scientific and public events, featured more than 500 talks, and strengthened collaboration across the UK astronomy community while showcasing Durham as an international hub for astronomy research.
- **Co-lead** of morning arXiv journal club at Durham University 2025
- **Member** of Faculty Candidate Interview Committee at KIAA, Postdoc Representative 2023, 2024
- **Organiser** of weekly Galaxy Party at KIAA 2023
- **Organiser** of the Postdoc Science Day at KIAA 2022
- **Organiser** of the Speaker Lunch at the Tsinghua Center for Astrophysics 2018-2019

INVITED SEMINAR TALKS

2026	DEEP-Theory meeting	University of California, Santa Cruz
	Astro Seminar at Kavli-IPMU	University of Tokyo
	Seminar of Galaxy and Cosmology	National Astronomical Observatories, CAS
	KIAA-DoA Seminar	Peking University
	Informal Talk	Tsinghua University
	Astronomy Seminar	Nanjing University
	Astronomy Seminar	Westlake University
2025	Astrophysics Research Institute Seminar	Liverpool John Moores University
	Lunchtime Talks at Astronomy Group	University of St Andrews
	Informal Talk	Tsinghua University
	KIAA-DoA Seminar	Peking University
	USTC Astronomy Seminar Series	University of Science and Technology of China
	DoA Seminar	Shanghai Jiao Tong University
2024	Galaxy & Cosmology seminar	Tsinghua University
	Lunch talk at South-Western Institute For Astronomy Research	Yunnan University
	Astronomy Seminar	Huazhong University of Science and Technology
	Carnegie arXiv Tea	Carnegie observatory
	CGI (Cosmology/Galaxies/IGM) Seminar	University of California, Santa Cruz
	KIPAC tea talk	Stanford University
2023	CGI (Cosmology/Galaxies/IGM) Seminar	University of California, Santa Cruz
	KIAA-DoA Seminar	Peking University
2022	Lunch Talk at the Department of Astronomy	Tsinghua University
2021	Lunch Talk at Kavli-IPMU	University of Tokyo

CONFERENCE TALKS

2026	DEX-XXII: 22nd Durham-Edinburgh eXtragalactic Workshop	Edinburgh
2025	SWIFTCON 2025	Leiden
	L-GALAXIES workshop 2025	Berlin
	National Astronomy Meeting (NAM) 2025	Durham
	Expanding the boundaries of dark matter halo	Shanghai
2024	Mock Barcelone 2024	Barcelone
	The Co-evolution of galactic eco-systems and their large-scale environments	Hangzhou
2023	The 2nd Shanghai Assembly on Cosmology and Structure Formation	Shanghai
	Collaboration Workshop on Cosmology and Galaxy Formation	Shanghai
	25th CAS Guoshoujing Symposium on Galaxies and Cosmology	Huangshan

	Conference of Star Formation and Nuclei Activity in Galaxies	Nanjing
2019	The 11-th Prime Focus Spectrograph collaboration meeting	Pasadena
2018	The 10-th Prime Focus Spectrograph collaboration meeting	Shanghai